

THE STRUCTURE OF MATHEMATICAL REALITY

Phase II — Chapter 4

Formal Systems and the Architecture of Proof

Note Template

Self-Study Curriculum

How to use this template

Complete these notes **after** the reading and the lecture. The goal is not to copy phrases from the text — it is to reconstruct the ideas in your own language. Leave every item blank on first pass; fill it in from memory, then check.

1. The three components of a formal system

- a.
- b.
- c.

2. A proof is a _____ procedure, not an _____.

3. Axioms are not _____, they are _____.

4. Godel's first incompleteness theorem says

your own words, do not use the word "incompleteness"

5. Godel's second incompleteness theorem says

your own words

6. What Godel does NOT mean

state at least two common misinterpretations

7. "Mathematics is not a single unified system", what it

actually is instead:

8. The Continuum Hypothesis is an example of

what kind of statement?

9. How this changes my understanding of mathematical certainty

one paragraph
